

Show Code

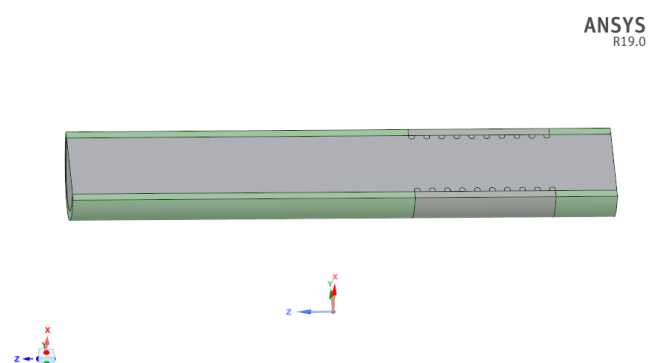
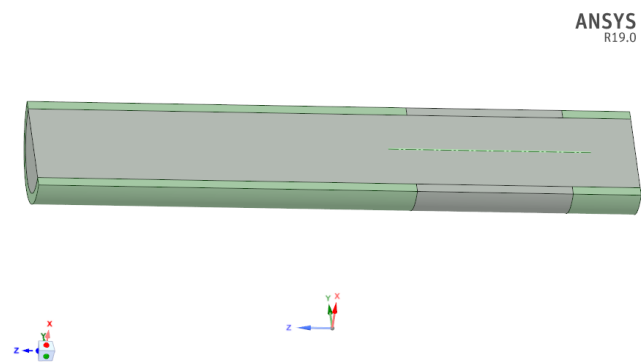
Analysis: Geometry - Coils vs No Coils

Comparing the airflow velocity between two geometrys: **coils** and **no coils**.

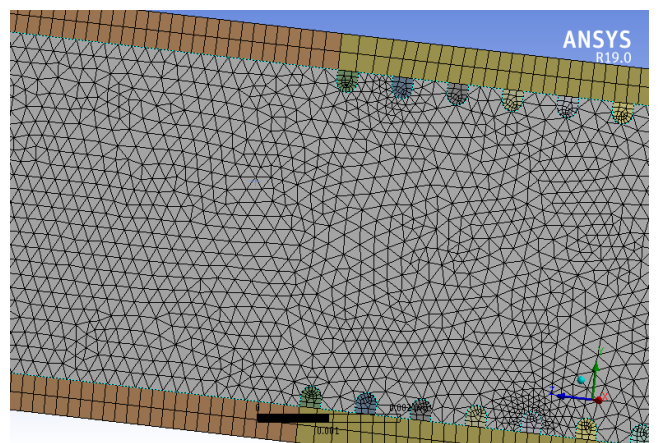
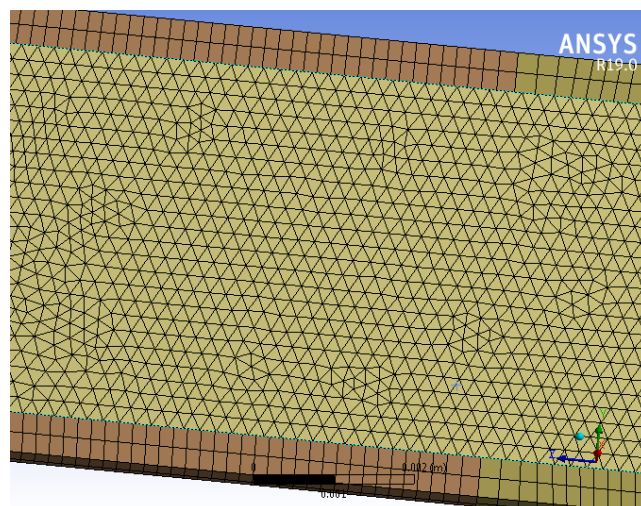
The comparison is at 6 points throughout the airflow volume.

- Point Bottom Line: Point close to the inlet
- Point Line at coil: Point at the first coil
- Point Line Between Coil: Point between coil 5 and coil 6
- Point at Line 1: Point between outlet and atomizer
- Point at Line 2: Point directly after the atomizer
- Point at Line 3: Point close to the outlet

Geometry



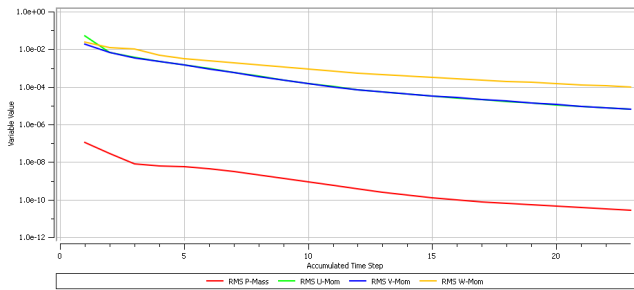
Mesh



CFX Solver Results

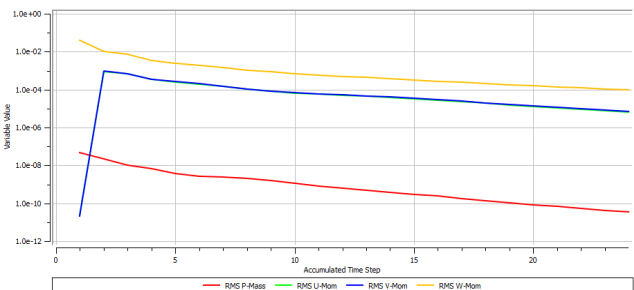
Momentum and Mass (No Coils)

Run Time = 36 seconds



Momentum and Mass (Coils)

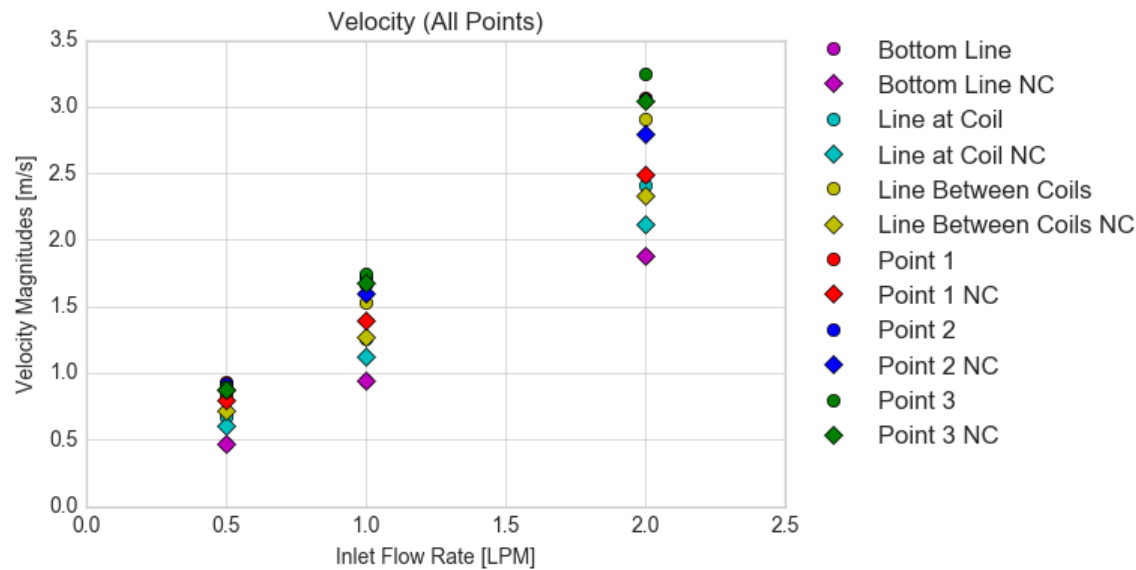
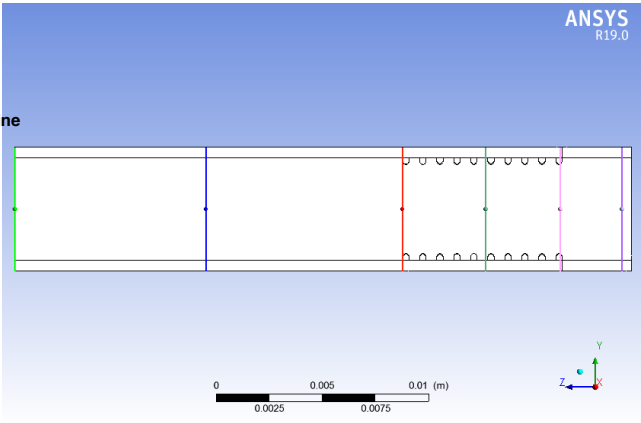
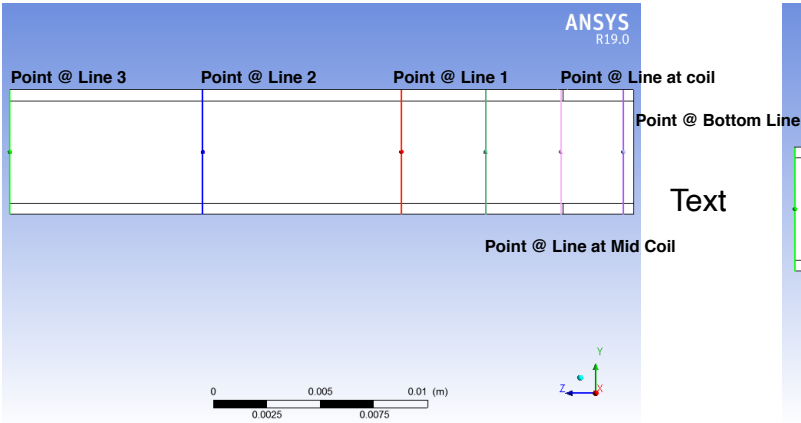
Run Time = 44 seconds

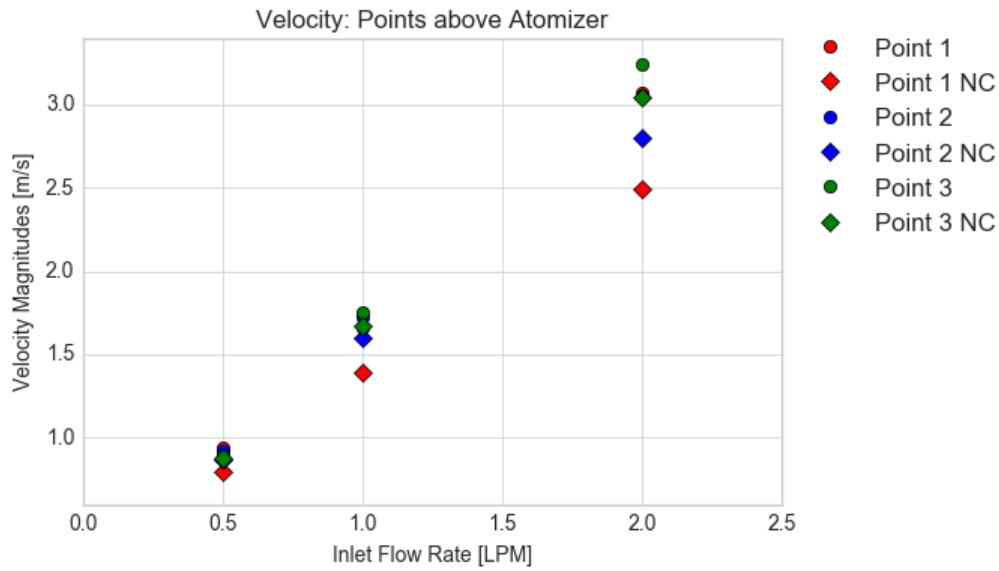
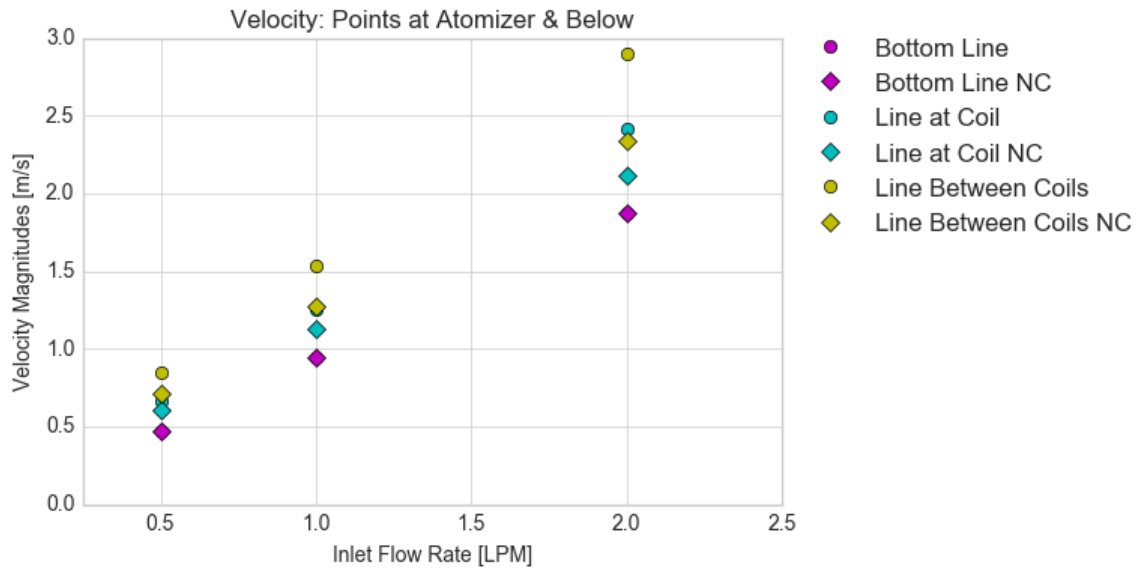


Results

Low Inlet Velocity

Inlet velocity is set at 0.4663 [m/s] or 0.5 [LPM]





Mesh Element Details

	No Coils	Coils	Difference	Percent Difference
Total Airflow Elements	262458	260270	2188	0.84067
Total Elements	270608	304013	33405	10.98802

Velocity Magnitudes Table (Low Inlet)

The Inlet Velocity is low at 0.4663 [m/s] or 0.5 [LPM]

Percent Difference is calculating: $(C - NC)/C$

	No Coils [m/s]	Coils [m/s]	Difference [m/s]	Percent Difference (%)
Point @ Bottom Line	0.47273034	0.47302601	0.0003	0.06251
Point @ Line at Coil	0.60700488	0.6668154	0.05981	8.96958
Point @ Line Between Coils	0.71369469	0.84512043	0.13143	15.55113
Point @ Line 1	0.79433727	0.93509191	0.14075	15.05249
Point @ Line 2	0.86949259	0.92357117	0.05408	5.85538
Point @ Line 3	0.87729371	0.89317906	0.01589	1.77852

Velocity Magnitudes Table (Medium Inlet)

The Inlet Velocity is medium at 0.9326 [m/s] or 1 [LPM]

Percent Difference is calculating: $(C - NC)/C$

	No Coils [m/s]	Coils [m/s]	Difference [m/s]	Percent Difference (%)
Point @ Bottom Line	0.9407618	0.94161326	0.00085	0.09043
Point @ Line at Coil	1.12465882	1.25512552	0.13047	10.39471
Point @ Line Between Coils	1.26961243	1.53492844	0.26532	17.28524
Point @ Line 1	1.38959277	1.66393948	0.27435	16.48778
Point @ Line 2	1.59824789	1.72159505	0.12335	7.1647
Point @ Line 3	1.67161703	1.74869156	0.07707	4.40755

Velocity Magnitudes Table (High Inlet)

The Inlet Velocity is high at 1.86531 [m/s] or 2 [LPM]

Percent Difference is calculating: $(C - NC)/C$

	No Coils [m/s]	Coils [m/s]	Difference [m/s]	Percent Difference (%)
Point @ Bottom Line	1.8743763	1.87662816	0.00225	0.12
Point @ Line at Coil	2.11138511	2.41516638	0.30378	12.57807
Point @ Line Between Coils	2.33466125	2.90170956	0.56705	19.54187
Point @ Line 1	2.49509096	3.06617975	0.57109	18.62542
Point @ Line 2	2.79569364	3.05551672	0.25982	8.50341
Point @ Line 3	3.0402801	3.24477029	0.20449	6.30215

Lines for Low Inlet

